

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Shipped 10+ production TypeScript/Node.js features: Pusher real-time event delivery pipelines, Temporal.io durable workflows, multi-model AI agent tooling (Claude/GPT-4o/Gemini), and Auth0 RBAC; 923 commits, 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

AlgoPulse	Agentic AI Platform — React Native + Real-Time Streaming	2025
- Architected React Native + Supabase Edge Functions agentic AI platform: streamed multi-step AI reasoning directly to mobile clients, replacing API polling and eliminating state loss at 2,000 concurrent async inference req/sec. - Scaled Temporal.io + Kafka worker fleet; sustained 2,000 req/sec with zero message loss across worker restarts.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard in TypeScript/NestJS: MySQL with idempotency-key deduplication and outbox pattern, MongoDB TTL projections, Redis sorted sets for sub-millisecond reads; SSE live push. - Validated multi-store consistency across MySQL, MongoDB, and Redis via Testcontainers integration tests against real containers in CI; benchmarked hot-path p99 latency with k6 and enforced gates in GitHub Actions.		
PerCent	GenAI Finance App — React Native	2024
- Built React Native cross-platform finance app end-to-end in a 48-hour sprint with goal tracking, expense logging, budget alerts, and an AI-powered financial assistant; ranked Top 10 of 100+ competing teams at HackNC 2024. - Built RAG chatbot on Qwen-14B-Chat with Hugging Face; Selenium-scraped financial data reduced manual research by 30%.		

Languages and Technologies

- Languages:** TypeScript, JavaScript, Python, Java, SQL, Kotlin
- Mobile & Real-Time:** React Native, Pusher, WebSockets, SSE, Supabase Edge Functions, cross-platform SDK patterns
- Backend:** Node.js, NestJS, FastAPI, Temporal.io, Kafka, PostgreSQL, Redis, MongoDB
- Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, OpenTelemetry, Grafana

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog